

EGCI 232 Engineering Electronics

Course Syllabus

Instructor: Dr. Somnida Bhatranand

Email: somnida.rat@mahidol.ac.th

Class Hours: Tuesday 12:00-13:50 /Friday 12:00-14:50

Classroom: R-232 and EE 401, Engineering Building 1

Course Description: Introduction to solid-state electronics as circuit elements, including diodes, bipolar transistors, rectifier circuits, voltage regulators, linear power supplies, power amplification, MOSFETs, applications in operational amplifiers in various feedback configurations.

Course Goals: To introduce fundamentals of solid-state devices and operation of basic electronic circuits. And students can describe and design basic electronic circuits for engineering applications.

Course Learning Outcomes: After completing this course students should be able to:

- understand the fundamentals of electronics within the fields of computer and electrical engineering.
- Understand diode circuits and models.
- Understand BJT and FET transistor circuits and models.
- Perform small-signal analysis for transistor circuits.
- Analyze amplifier circuits.
- Know how to use and analyze operational amplifiers.
- Know how to design and build electronic circuits virtually.

References:

- Donald A. Neamen, "Electronic Circuit Analysis and Design".
- Thomas L. Floyd, "Electronic Devices", 8th Ed., Pearson Higher Education.
- Robert L. Boylestad, "Electronic Devices and Circuit Theory" 10th Ed., Pearson Higher Education.

Virtual Lab Website: www.tinkercad.com

Scores:

2 Quizzes	10%
Midterm	25%
Final	25%
Lab skills	15%
Class project	15%
Assignments	10%

EGCI 232 Engineering Electronics
Course Syllabus

Tentative Schedule:

Date		Topics	Room
Tuesday	15/9	Introduction to the class and ECA review	6276
Friday	18/9	Semiconductors and PN Junction Diode	6276
Tuesday	22/9	Lab # 1 – Introduction to Electronics Laboratory	EE 401
Friday	25/9	Diode Circuits and applications	6276
Tuesday	29/9	Lab # 2 – Diode Circuits	EE 401
Friday	2/10	Bipolar Junction Transistor (BJT)	6276
<i>Tuesday</i>	<i>6/10</i>	<i>Commencement Ceremony</i>	<i>No class</i>
Friday	9/10	BJT biasing techniques	6276
Saturday	10/10	Simulation tools for Electronics (1:00 – 2:00 pm = 2 hours)	6276
<i>Tuesday</i>	<i>13/10</i>	<i>National Holiday</i>	<i>No class</i>
Friday	16/10	Common Emitter Amplifiers	6276
Saturday	17/10	Other BJT Amplifiers (1:00 – 2:00 pm = 2 hours)	6276
Tuesday	20/10	Lab # 3 – Common Emitter Amplifiers	EE 401
<i>Friday</i>	<i>23/10</i>	<i>National Holiday</i>	<i>No class</i>
Tuesday	27/10	Quiz #1	6276
Friday	30/10	Midterm Exam	6276
Tuesday	3/11	MOSFET Biasing	6276
Friday	6/11	MOSFET Amplifiers	6276
Tuesday	10/11	Lab # 4 – MOSFET Amplifiers	EE 401
Friday	13/11	Multistage amplifier and current source	6276
Tuesday	17/11	Differential amplifier	6276
Friday	20/11	Operational Amplifiers	6276
Tuesday	24/11	Lab # 5 - OP-AMP Circuits	EE 401
Friday	27/11	Quiz #2	6276
Tuesday	1/12	Group 1 Class project	EE 202
Friday	4/12	Group 2 Class project	EE 202
Sunday	6/12	Review for Final Exam (9.00-11.50 am = 3 hours)	6276
Tuesday	8/12	Final Exam (2 hrs.)	TBA

EGCI 232 Engineering Electronics

Course Syllabus

Grading Policy: Grades for the course determined from the following scale:

90% - 100%	A	70% - 74%	C
85% - 89%	B+	65% - 69%	D+
80% - 84%	B	60% - 64%	D
75% - 79%	C+	0% - 59%	F